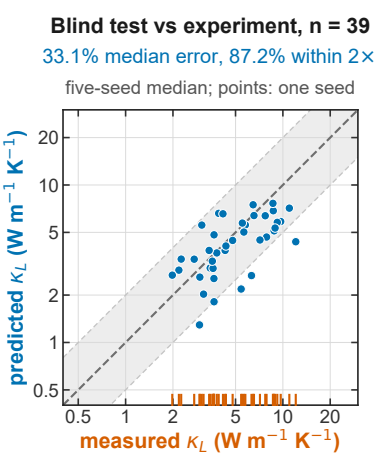

Crystal structure
half Heusler XYZ
cubic $C1_b$, VEC 18
Machine learning
CatBoost, 45 descriptors
vs DFT: 21.9% median
cluster held out
κ_L^{BTE}
Transfer function
$\kappa_L^{\mathrm{exp}} = \kappa_L^{\mathrm{BTE}} \times$
$\min[c\,(T/300)^p, 1]$
$c = 0.42,\ p = 0.70$
κ_L^{exp}


Blind test vs experiment, n = 39
33.1% median error, 87.2% within 2×
five-seed median; points: one seed
predicted κ_L (W m^{-1} K^{-1})
measured κ_L (W m^{-1} K^{-1})
0.5
1
2
5
10
20